One picture, several answers

What multi-label reading does, and how it becomes the edge algorithm

Weekly review, Monday 9 November 2026

All numbers are measured on the annotation of record. The population is named on every table.

The record now covers all 24 production files

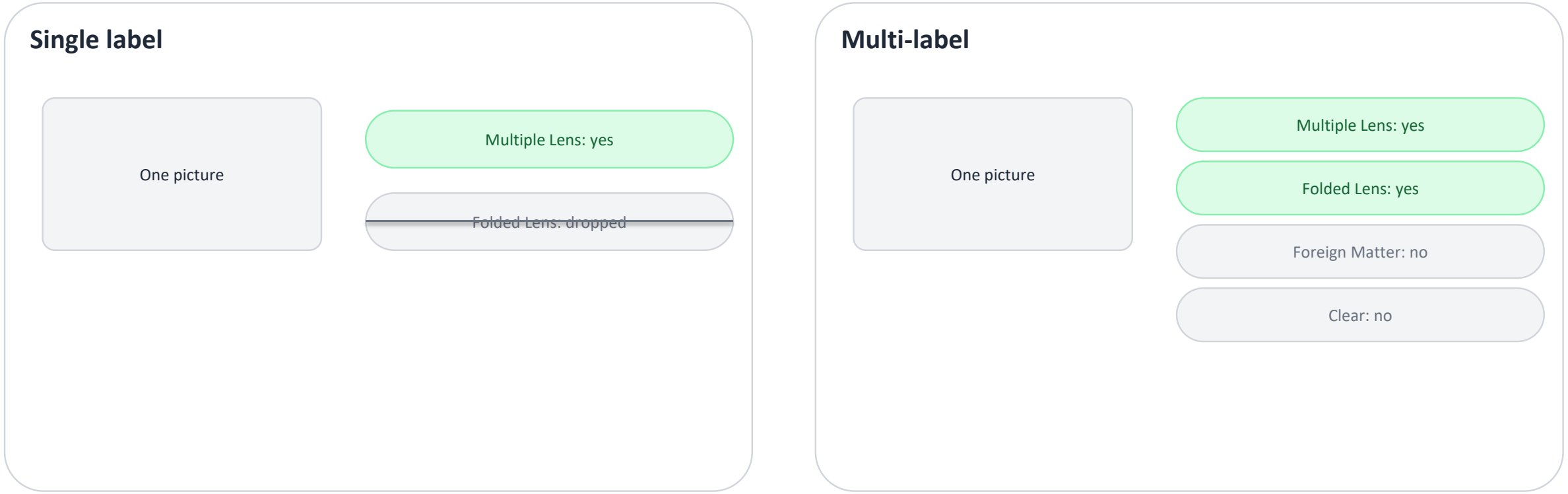
17 new files and 7 revised files. The older libraries are unchanged.

Measure	Now
Pictures examined	10,614
Pictures with a mark	5,500
Clear pictures	2,117
Pictures in the labelled record	7,617
Object marks	37,199 on 2,332 pictures
Production files at the top of the record	24

- The 24 files hold 3,597 picture entries. 719 of those pictures are new to the record; the rest repeat pictures already there.
- The revised files changed 133 of their 1,047 pictures. Most changes merged three bubble labels into one.
- The experiment uses 7,470 of the 7,617 labelled pictures. The 147 left out carry a retired class or no usable object label.

One picture can carry more than one defect

The guide records each condition separately. A single-label model would have to choose one and drop the rest.



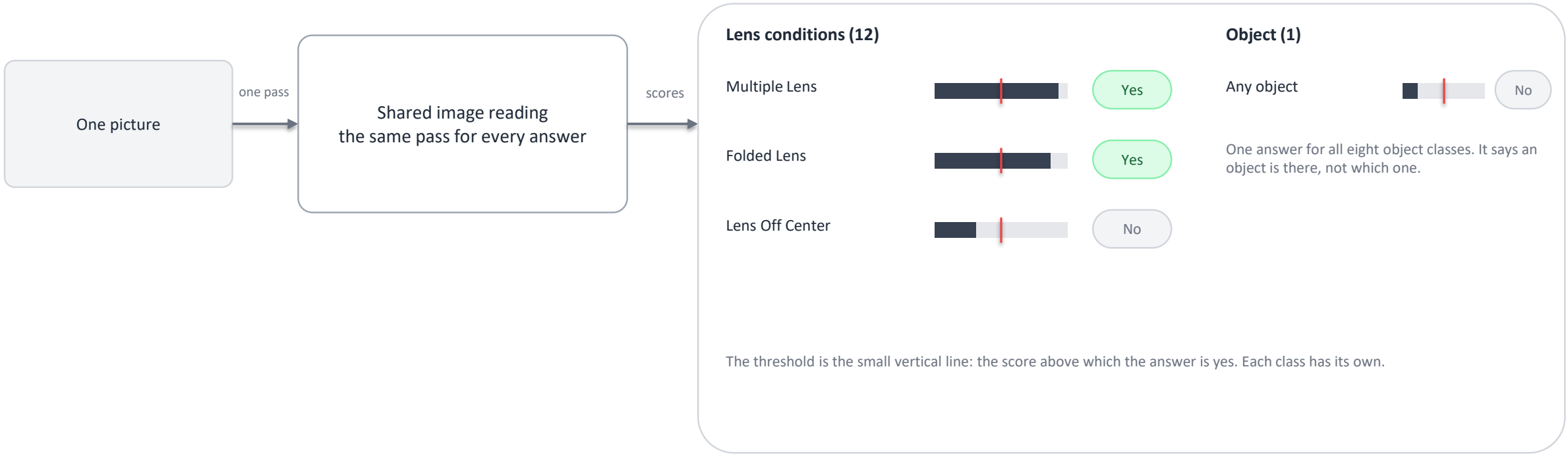
2,739 of the 7,470 pictures carry two or more answers.

1,524 carry both a lens condition and an object.

Almost every folded lens picture is also a multiple lens picture. A single label drops one of the two.

One pass, an answer for each class

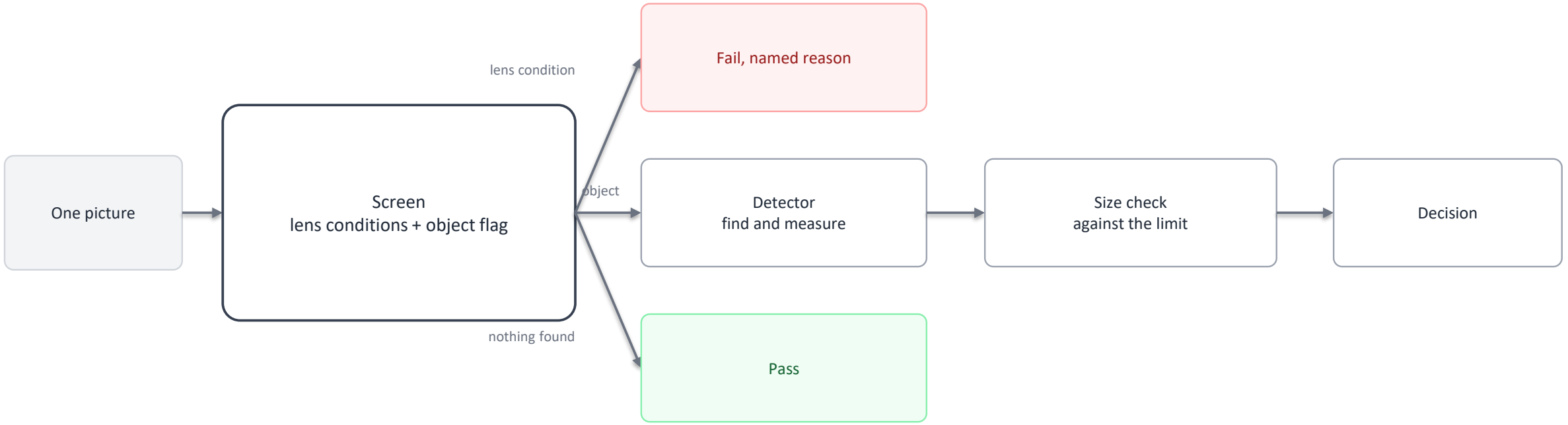
Twelve lens conditions and one object flag. Thirteen answers, each with its own threshold.



- Each answer is a score from 0 to 1, not a choice between classes.
- The object flag is one answer for all eight object classes.
- The model does not need a separate pass for each question.

The screen decides the route. The detector does the measuring.

This is where the multi-label model sits in the edge check.



- The screen answers two questions in one pass: is the lens or package wrong, and is there an object at all.
- Only the detector measures size. The model decides which pictures need it.
- 2,332 of the 7,617 labelled pictures carry an object mark. The detector would see those, not all of them.

Bubbles take the pass route unless a lens condition is also present.

Pictures per class: train, validation, test

One frozen split of 7,470 pictures. A picture can appear in more than one row.

Class	Train	Validation	Test	All
Missing Primary Package	1,000	214	214	1,428
Multiple Lens	998	214	214	1,426
Lens Off Center	841	180	180	1,201
Dirty Camera	401	86	86	573
No Lens	228	49	49	326
Folded Lens	249	33	51	333
Low Dose	474	101	102	677
Cavity Off Center	129	28	28	185
Dirty Strobe	23	8	2	33
View Obstructed	44	9	10	63
Bubble Scatter	3	1	1	5
Package Misalignment	23	5	5	33
Any object (one flag)	1,563	341	343	2,247
Clear pictures	1,482	318	317	2,117
All pictures	5,239	1,110	1,121	7,470

Train: used to fit the model. Validation: used only to choose the thresholds. Test: read once, at the end.

Counts are pictures carrying the class, not the size-flagged subset. The eight object classes are one flag in this run.

The two runs, on all 7,470 pictures

The October design, remeasured on the three-way split. The second run adds one object flag.

	Lens only	Lens + object flag
Answers per picture	12	13
Lens AUPRC (8 classes)	0.836	0.818
Macro F1 at 0.5	0.722	0.724
Macro F1 at chosen thresholds	0.801	0.781
Exact match at 0.5	0.789	0.764
Exact match at chosen thresholds	0.777	0.762
Inference time per picture	5.76 ms	5.59 ms

5,239
train

1,110
validation

1,121
test

Both runs read the same 7,470 pictures.

- AUPRC is average precision and does not need a threshold. F1 balances misses and false alarms at a threshold. 1 is perfect on both.
- The chosen thresholds raise macro F1 but not the exact-match count. A picture is an exact match only when all twelve lens answers are right.
- The October readout used a two-way split and chose thresholds on the test split. These runs use a three-way split and choose thresholds on validation, so the new scores replace the old ones.

The object flag is accurate enough to route

Measured on the 1,121 test pictures, at the default threshold and at the threshold chosen on validation.

Group	At 0.5	At chosen threshold	Target
Object, no lens condition	118 of 120 (98.3%)	117 of 120 (97.5%)	at least 90%
Lens condition and object	212 of 223 (95.1%)	211 of 223 (94.6%)	none
Clear	1 of 317 (0.3%)	0 of 317	at most 5%

- A miss sends a dirty picture down the pass route. A false alarm sends a good picture to the detector. Both are counted here.
- The chosen threshold is 0.55. The numbers barely move from the default, so the flag is not sensitive to the threshold.
- The flag covers all eight object classes together. It says an object is there, not which one.

The threshold is chosen on validation, not on test

The threshold is part of the method, not a setting copied from the last data set.



- Choose on validation pictures the model never trained on. Score the test pictures once, after the thresholds are frozen.
- Pick the threshold with the highest F1 on validation. A class with fewer than 10 validation positives keeps 0.5.
- Thresholds do not carry over between data sets. The full-data model gets its own.
- For routing, only the object flag's threshold matters. The per-class thresholds apply to the lens answers.

0.5 is the default threshold. The chosen threshold is picked for each class on validation.

The threshold each class uses

The lens-plus-flag run. Four classes keep 0.5 because they have fewer than 10 validation positives.

Class	Validation	Test	Threshold	Basis
Missing Primary Package	214	214	0.25	Chosen on validation
Multiple Lens	214	214	0.60	Chosen on validation
Lens Off Center	180	180	0.45	Chosen on validation
Dirty Camera	86	86	0.30	Chosen on validation
No Lens	49	49	0.15	Chosen on validation
Folded Lens	33	51	0.45	Chosen on validation
Low Dose	101	102	0.40	Chosen on validation
Cavity Off Center	28	28	0.30	Chosen on validation
Dirty Strobe	8	2	0.50	Default, under 10
View Obstructed	9	10	0.50	Default, under 10
Bubble Scatter	1	1	0.50	Default, under 10
Package Misalignment	5	5	0.50	Default, under 10
Any object (one flag)	341	343	0.55	Chosen on validation

The threshold is chosen on validation by the highest F1. A class with fewer than 10 validation positives keeps 0.5 and is marked Default. The test count is shown beside it.
Clear has no threshold; it is the all-zero row. Surface Groove has no pictures in this data.

What we learned

- **The lens job held.** The lens AUPRC is 0.836 with twelve answers and 0.818 with thirteen. Dirty Camera accounts for most of the difference.
- **The flag routes.** 97.5 percent of object-only pictures at the chosen threshold. No clear picture fired.
- **The screen is fast.** 5.6 to 5.8 ms per picture on the A100, measured over 1,121 pictures.
- **Folded Lens is still weak.** 0.42 on its own and 0.49 with the flag, under the 0.70 target.

The routing idea holds: the screen decides, the detector measures, and the size check stays.

What we do next

1. **Split the object flag into the eight classes.** Every fail then has a name, not just a flag.
2. **Connect the detector.** Object location, size in microns, and the size check.
3. **Collect more folded lens pictures.** The ones that are not also multiple lens.
4. **Run both again with a second starting point.** So a small difference can be told from noise.

The screen is the first stage of the edge algorithm. The detector and the size check are the second.